DATA ENGINEER

Interview-Ready List of Questions **_Experience: 3-5 Year_**

CORE DATA ENGINEERING CONCEPTS

1. Explain end-to-end data pipeline architecture you worked on.
2. Difference between ETL and ELT. When would you choose each?
3. What is schema evolution and how do you handle it?
4. How do you design a scalable data ingestion framework?
5. What is idempotency in data pipelines and why is it important?
6. How do you handle late-arriving data?
7. Explain batch vs streaming data processing with use cases.
8. What is data lineage and why does it matter?
9. How do you ensure data quality in pipelines?
10. How do you handle data reconciliation issues?

SQL & DATA MODELING

11. How do you find duplicate records and remove them using SQL?
12. Write a query to get the Nth highest salary.
13. Difference between INNER JOIN, LEFT JOIN, and FULL JOIN.
14. Explain window functions and real-time use cases.
15. What are slowly changing dimensions (SCD)?
16. Difference between star schema and snowflake schema.
17. How do you optimize a slow-running SQL query?
18. What is partitioning and bucketing?
19. How do indexes work internally?
20. Difference between OLTP and OLAP systems.

BIG DATA & SPARK

21. Explain Spark architecture.
22. Difference between RDD, DataFrame, and Dataset.
23. What is data skew and how do you handle it?
24. Explain shuffle in Spark.
25. What are wide and narrow transformations?
26. Difference between cache and persist.
27. What causes OutOfMemory errors in Spark?

28. How do you optimize Spark jobs?
29. What is Z-ordering?
30. Difference between Parquet and ORC.

CLOUD & DATA PLATFORMS (AWS / AZURE / GCP)

31. Explain your cloud data architecture.
32. Difference between Data Lake and Data Warehouse.
33. How does Azure Data Factory / AWS Glue work?
34. What is CDC and how do you implement it?
35. How do you handle incremental data loads?
36. What is Delta Lake and why is it used?
37. Explain Medallion Architecture.
38. How do you secure data in cloud platforms?
39. Difference between managed and external tables.
40. How do you handle cost optimization in cloud data systems?

STREAMING & REAL-TIME PROCESSING

41. Explain Kafka architecture.
42. Difference between Kafka and traditional messaging systems.
43. What is consumer group in Kafka?
44. How do you ensure exactly-once processing?
45. Difference between event time and processing time.
46. What is watermarking?
47. How do you handle message reprocessing?
48. What happens if a Kafka broker goes down?
49. How do you manage schema changes in streaming?
50. Batch vs Lambda vs Kappa architecture.

ORCHESTRATION & WORKFLOW

51. How does Apache Airflow work?
52. Difference between DAG, Task, and Operator.
53. How do you handle task failures?
54. How do you implement retry and alert mechanisms?
55. How do you manage dependencies between jobs?
56. How do you backfill historical data?
57. What is SLA in pipelines?
58. How do you monitor data pipelines?
59. Difference between scheduling and orchestration.
60. How do you design fault-tolerant pipelines?

DATA GOVERNANCE & SECURITY

61. How do you implement data masking?
62. Difference between encryption at rest and in transit.
63. What is RBAC?

64. How do you manage PII data?
65. What is GDPR compliance from a data engineering perspective?
66. How do you track data access?
67. What is metadata management?
68. How do you handle audit requirements?
69. What is data catalog?
70. How do you manage schema registry?

DEVOPS & PRODUCTION

71. How do you version control data pipelines?
72. CI/CD for data engineering – how do you implement it?
73. How do you test data pipelines?
74. Difference between unit test and data validation test.
75. How do you deploy changes safely to production?
76. How do you handle rollback in pipelines?
77. What metrics do you monitor in production?
78. How do you debug pipeline failures?
79. How do you manage configuration across environments?
80. How do you document pipelines?

SCENARIO-BASED / REAL INTERVIEW QUESTIONS

81. Source system sends duplicate data — how will you handle it?
82. Pipeline is slow only on month-end data — what will you check?
83. Spark job works in dev but fails in prod — why?
84. Data count mismatch between source and target — steps to debug?
85. How do you reprocess data for a past date?
86. Pipeline failed midway — how do you recover?
87. Business wants near real-time dashboards — what will you design?
88. How do you handle sudden spike in data volume?
89. Schema changed in source without notice — what will you do?
90. How do you migrate on-prem data to cloud?

BEHAVIORAL / EXPERIENCE-BASED

91. Explain a challenging data issue you solved.
92. How do you prioritize pipeline issues?
93. How do you handle tight delivery timelines?
94. Describe a production incident you handled.
95. How do you collaborate with analytics and business teams?
96. How do you handle ambiguous requirements?
97. What trade-offs did you make in your architecture?
98. How do you keep yourself updated?
99. What improvements did you bring to your current project?
100. Why should we hire you as a Data Engineer?